

The December 1998 release of Java was called Java 2, and the version name J2SE 1.2 was given to JDK 1.2 to distinguish it from the other platforms of Java. Again, J2SE 1.5 (JDK 1.5) was renamed J2SE 5.0 and later as Java SE 5, citing the maturity of J2SE over the years as the reason for this name change. The latest version of Java is Java SE 9, which was released in September 2017. But actually, when you say Java 9, you mean JDK 1.9. So, keep in mind that Java SE was formerly known as Java 2 Platform, Standard Edition or J2SE.

The Java Development Kit (JDK) is an implementation of one of the Java Platforms, Standard Edition, Enterprise Edition, or Micro Edition in the form of a binary product. The JDK includes the JVM and a few other tools like the compiler (javac), debugger (jdb), applet viewer, etc, which are required for the development of Java applications and applets. The latest version of JDK is JDK 9.0.1 released in October 2017. OpenJDK is a free and open source implementation of Java SE. The OpenJDK implementation is licensed under the GNU General Public License (GNU GPL). The Java Class Library (JCL) is a set of dynamically loadable libraries that Java applications can call at run time. JCL contains a number of packages, and each of them contains a number of classes to provide various functionalities. Some of the packages in JCL include `java.lang`, `java.io`, `java.net`, `java.util`, etc.

The 'Hello World' program in Java

Other than console based Java application programs, special classes like the applet, servlet, swing, etc, are used to develop Java programs to complete a variety of tasks. For example, Java applets are programs that are embedded in other applications, typically in a Web page displayed in a browser. Regular expressions can be used in Java application programs and programs based on other classes like the applet, swing, servlet, etc, without making any changes. Since there is no difference in the use of regular expressions, all our discussions are based on simple Java application programs. But before exploring Java programs using regular expressions let us build our muscles by executing a simple 'Hello World' program in Java. The code given below shows the program `HelloWorld.java`.

```
public class HelloWorld
{
    public static void main(String[] args)
    {
        System.out.println("Hello World");
    }
}
```

To execute the Java source file `HelloWorld.java` open a terminal in the same directory containing the file and execute the command:

```
javac HelloWorld.java.
```

Now a Java class file called `HelloWorld.class` containing the Java bytecode is created in the directory. The JVM can be invoked to execute this class file containing bytecode with the command:

```
java HelloWorld.class
```

The message 'Hello World' is displayed on the terminal. Figure 2 shows the execution and output of the Java program `HelloWorld.java`. The program contains a special method named `main()`, the starting point of this program, which will be identified and executed by the JVM. Remember that a method in an object oriented programming paradigm is nothing but a function in a procedural programming paradigm. The `main()` method contains the following line of code, which prints the message 'Hello World' on the terminal:

```
'System.out.println("Hello World");'
```

The program `HelloWorld.java` and all the other programs discussed in this article can be downloaded from opensourceforu.com/article_source_code/January18javaforu.zip.

```
# javac HelloWorld.java
# java HelloWorld
Hello World
```

Figure 2: `Hello World` program in Java

Regular expressions in Java

Now coming down to business, let us discuss regular expressions in Java. The first question to be answered is 'What flavour of regular expression is being used in Java?' Well, Java uses PCRE (Perl Compatible Regular Expressions). So, all the regular expressions we have developed in the previous articles describing regular expressions in Python, Perl and PHP will work in Java without any modifications, because Python, Perl and PHP also use the PCRE flavour of regular expressions.

Since we have already covered much of the syntax of PCRE in the previous articles on Python, Perl and PHP, I am not going to reintroduce them here. But I would like to point out a few minor differences between the classic PCRE and the PCRE standard tailor-made for Java. For example, the regular expressions in Java lack the embedded comment syntax available in programming languages like Perl. Another difference is regarding the quantifiers used in regular expressions in Java and other PCRE based programming languages. Quantifiers allow you to specify the number of occurrences of a character to match against a string. Almost all the PCRE flavours have a greedy quantifier and a reluctant quantifier. In addition to these two, the regular expression syntax of Java has a possessive quantifier also.